

L1 - Queues

Special class

Queue

LIFO

MacBook

first come
first serve

type of DS

FIFO

first in

pop

MB



0
1
2

0
1
3



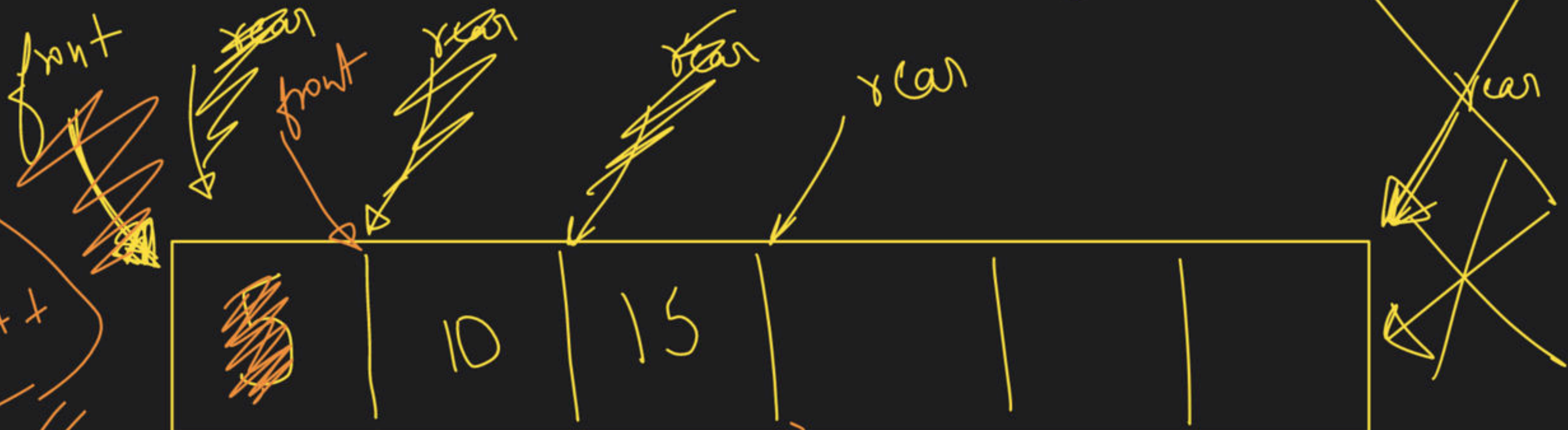
Queue

push $\rightarrow$ rear ++

Empty queue $\rightarrow$

front == rear ✓

pop $\rightarrow$ front ++



Queue <int>

push(5)

push(10)

push(15)

pop()

Types

normal queue

push

rear

pushback

pop

front

pop front

i/p restricted queue

o/p restricted queue

Circular queue

deque

push

front

rear

pop

front

rear

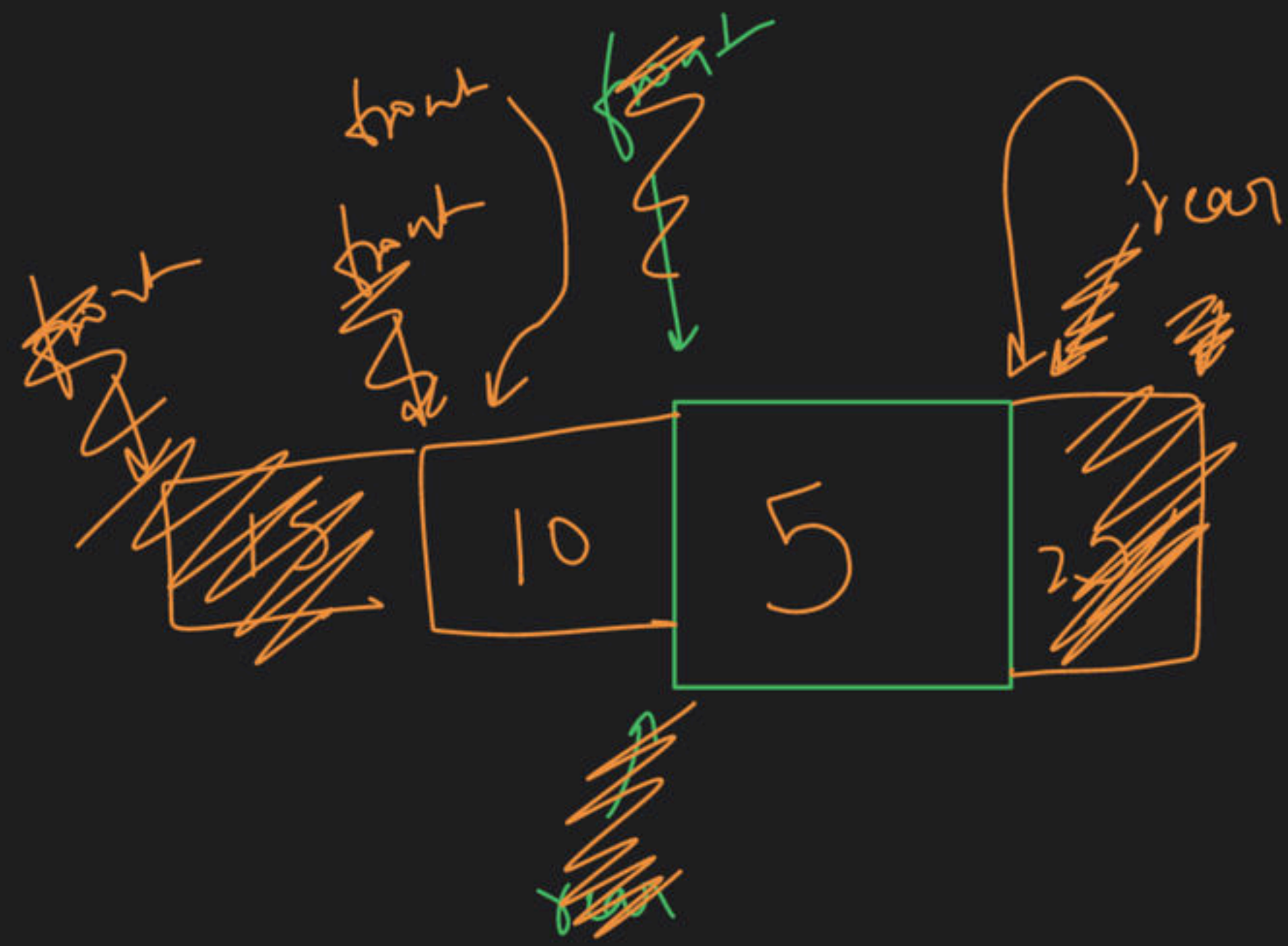


→ C++ STL

containers →

queue STL ✓
complete

deque<int> dq;



dq.pushfront(5)

dq.pushfront(10)

dq.pushfront(15)

dq.pushback(25)

dq.popback()

dq.popfront()

Normal Queue

LL $\rightarrow$ Node $\rightarrow$ data pointer

~~queue(int)~~

~~queue(int)~~

Scratch

class

stack

arr
size
top

-1

if not
top++
arr[top] = value

Queue

Custom
data
type

arr
==
size
front = -1
rear = -1

✓
array

LL

Owner



Class



members



all



size

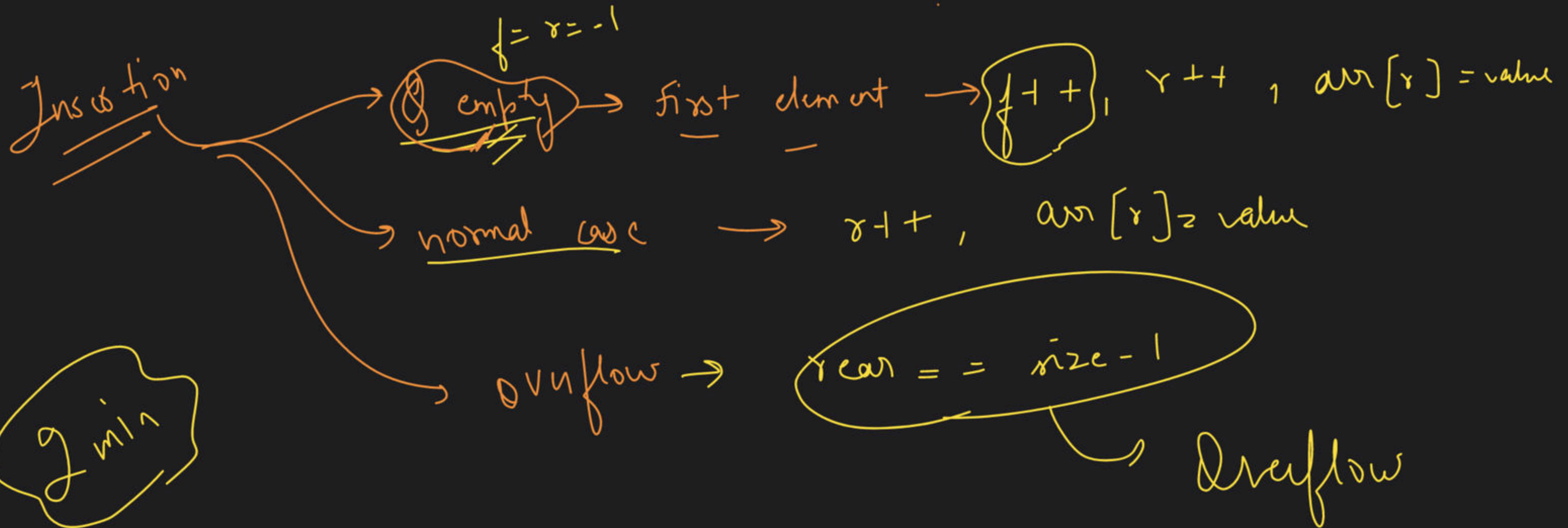
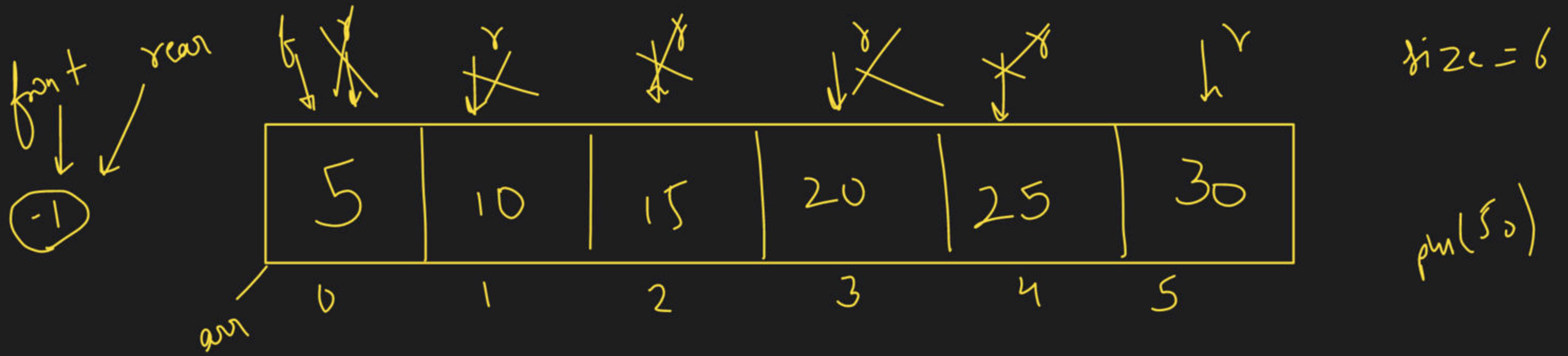


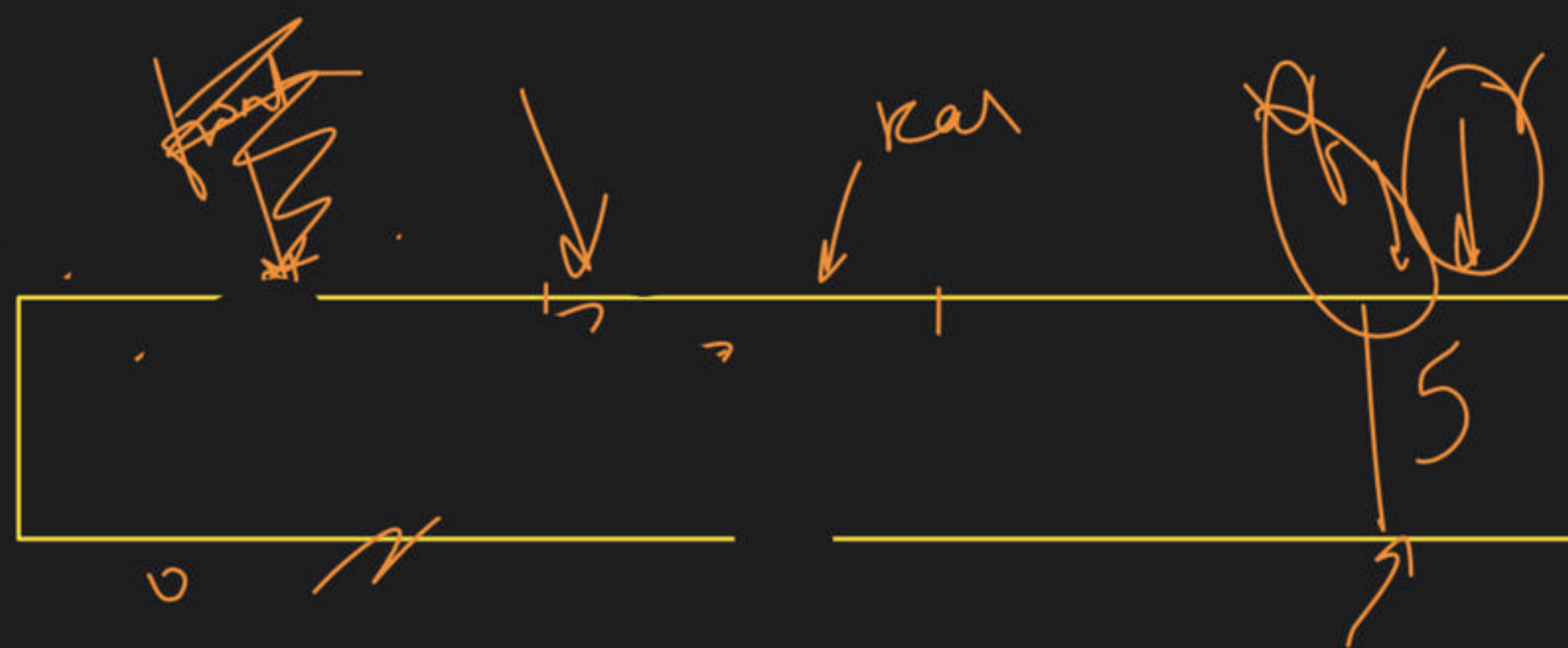
font



year

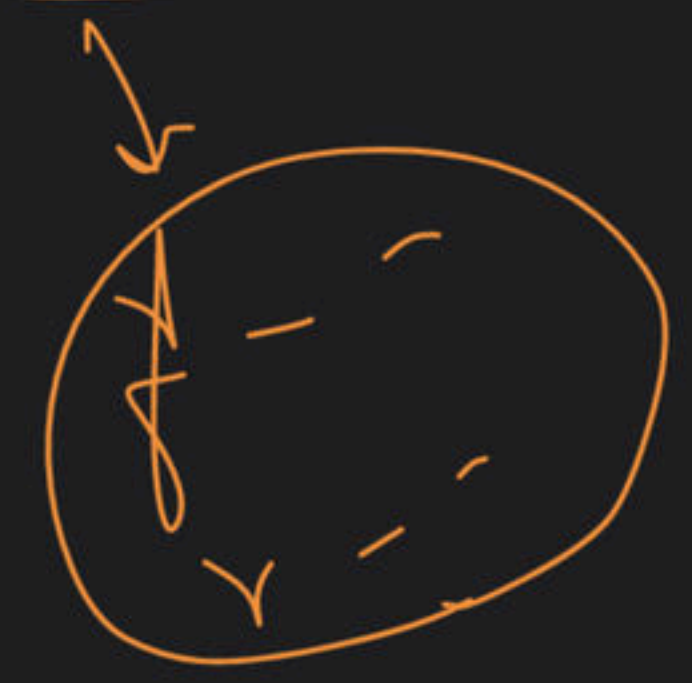






$f = -1, r = -1$
empty

$f = r$ but $\neq -1$



pop or Removal

Empty queue $\rightarrow$ Underflow

Single element $\rightarrow$

$f \rightarrow r \rightarrow -1$

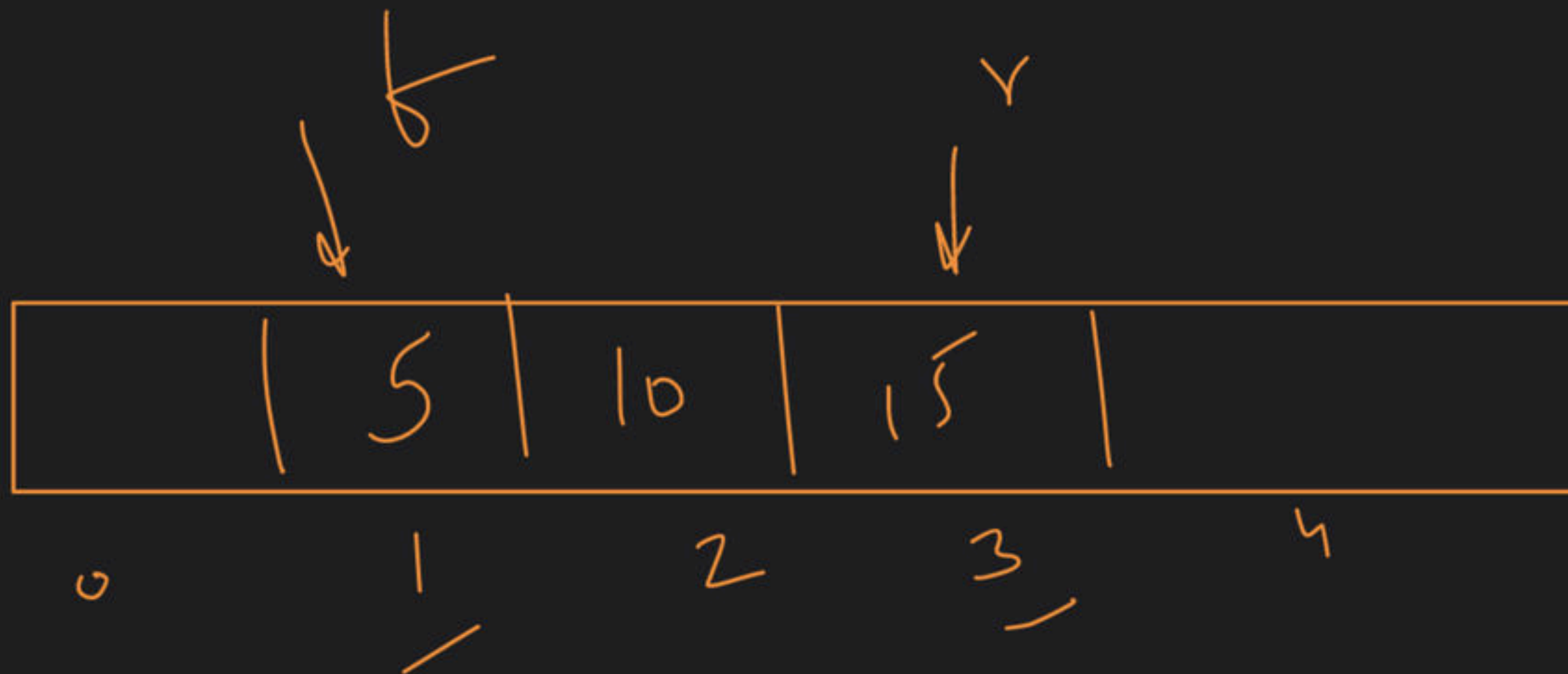
$f == r$

but $\neq -1$

normal pop -

$front++$

corner case?



$$\text{size} = r - f + 1$$

Custom



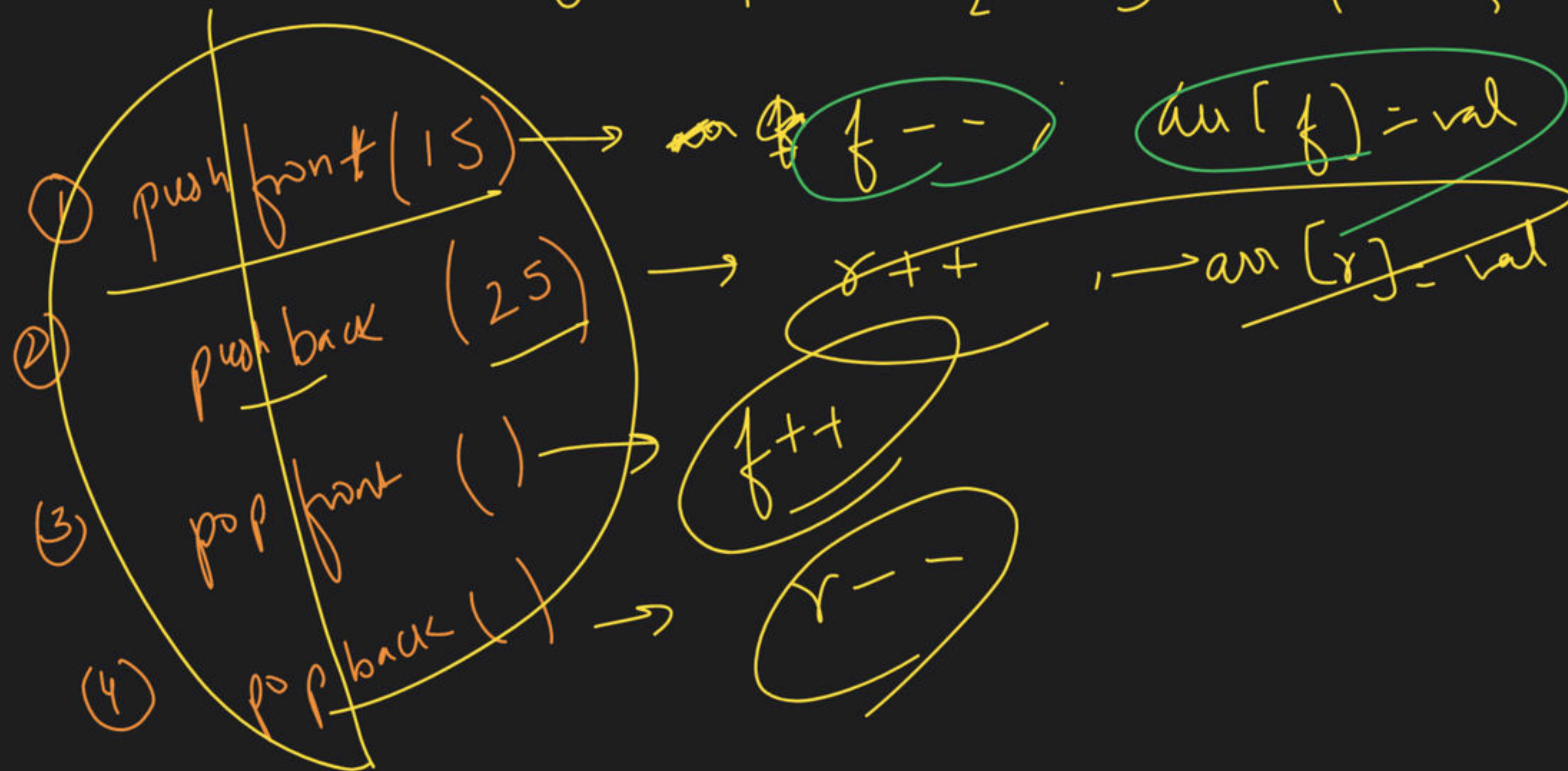
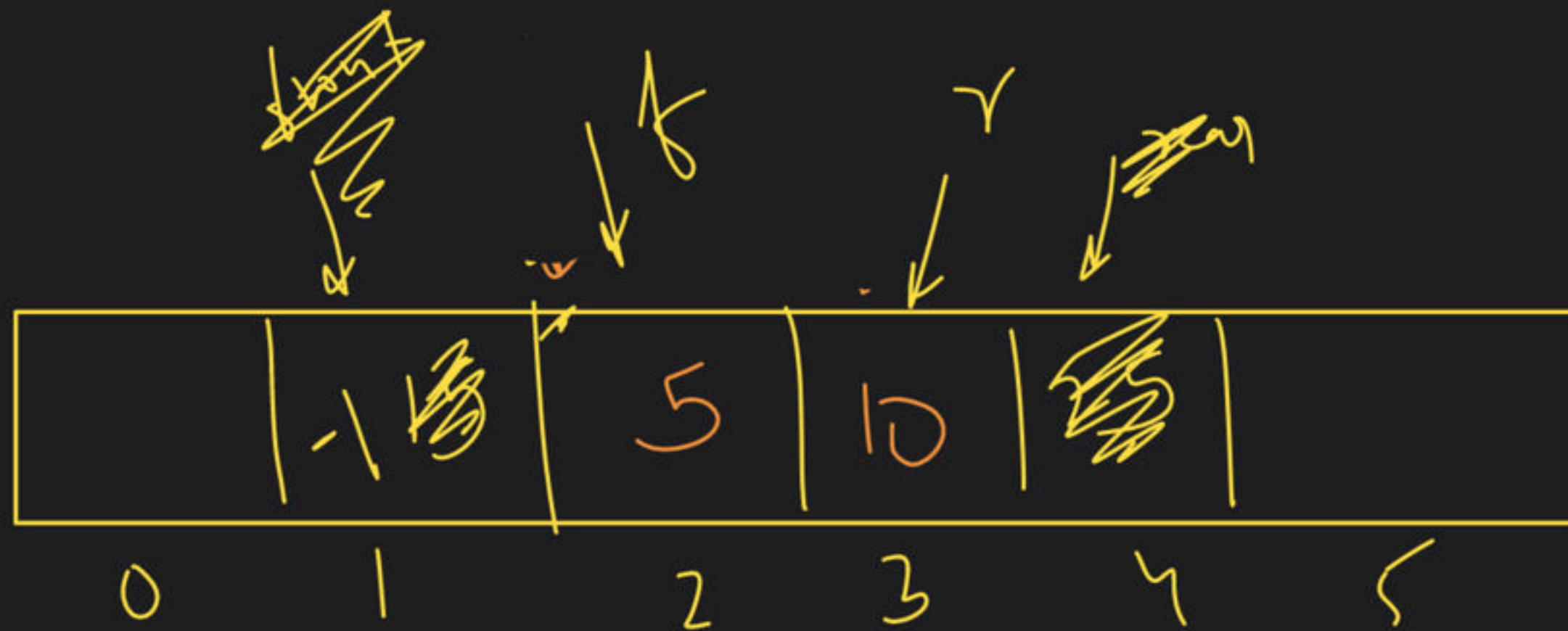
Normal

Image

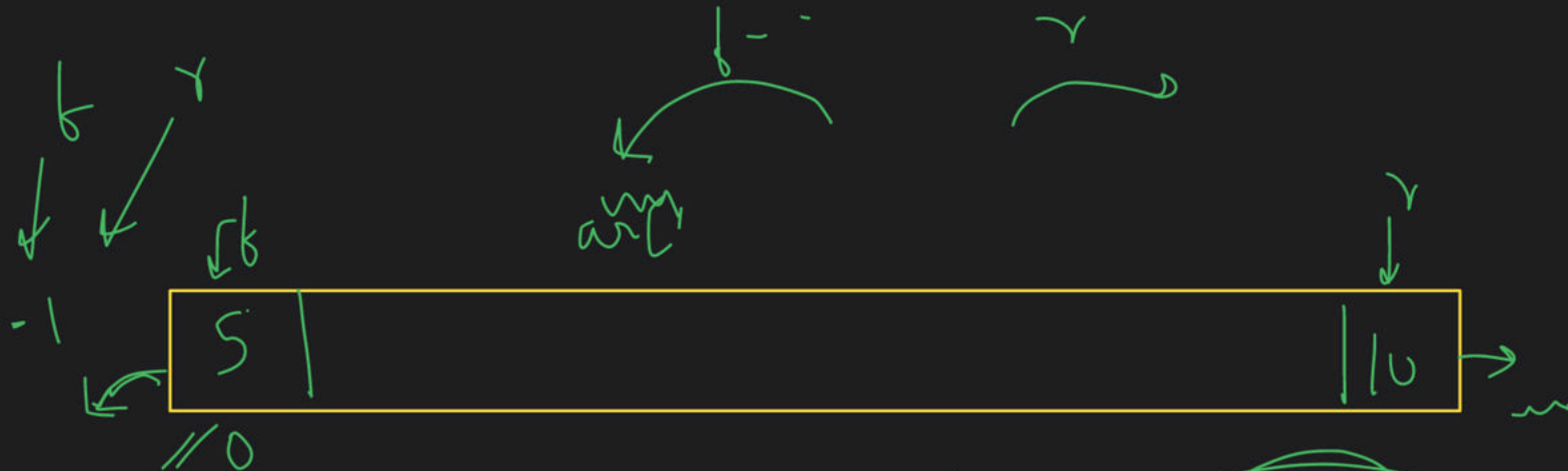
Custom



Degree



1) Deque



push front()

overflow

f == 0

first element /
if empty

normal $\rightarrow f--$, arr[f] = val

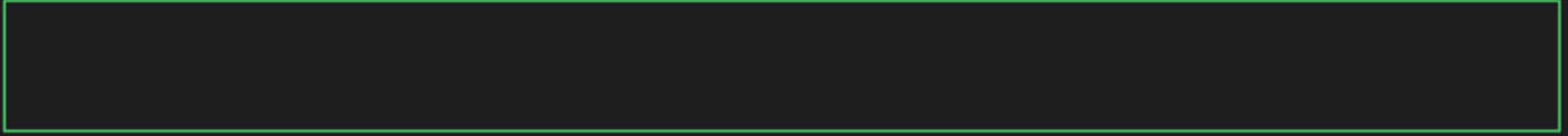
push back

overflow

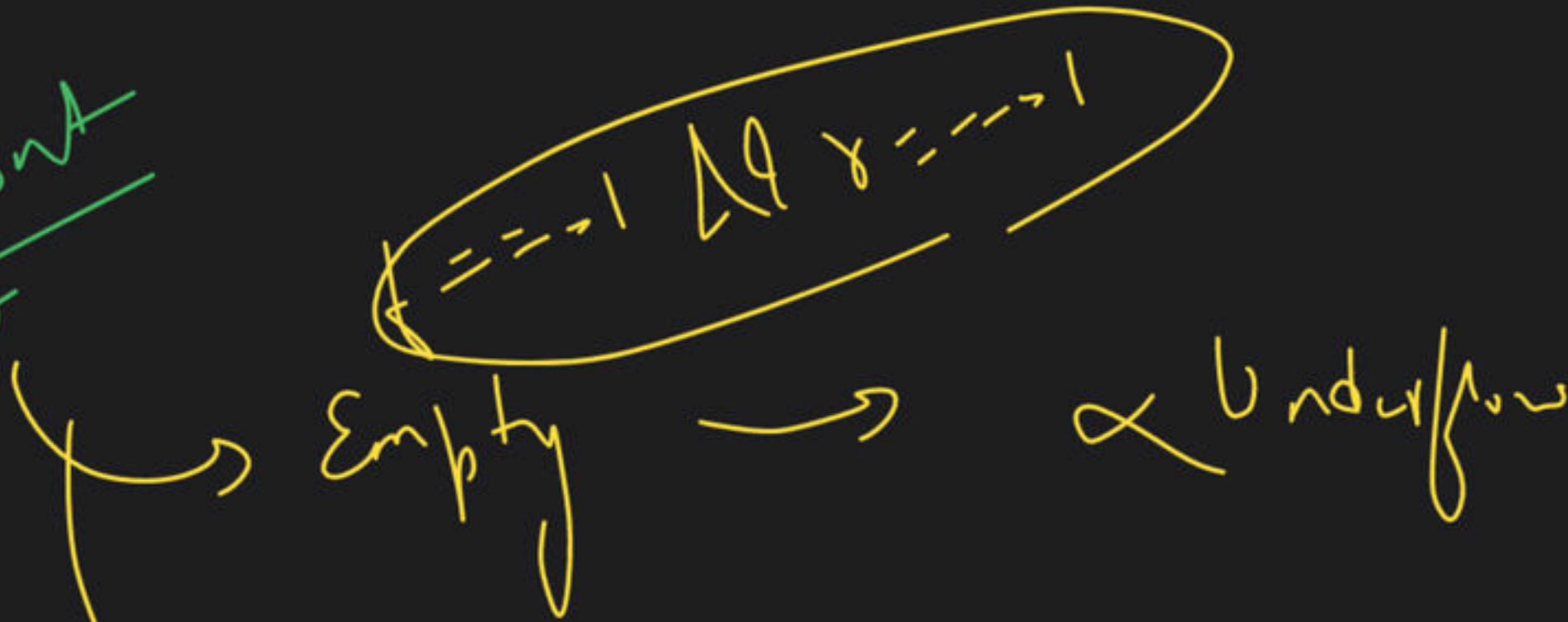
r = size - 1

$\rightarrow$ ~~~~~

r++, arr[r] = val



pop front



pop back()



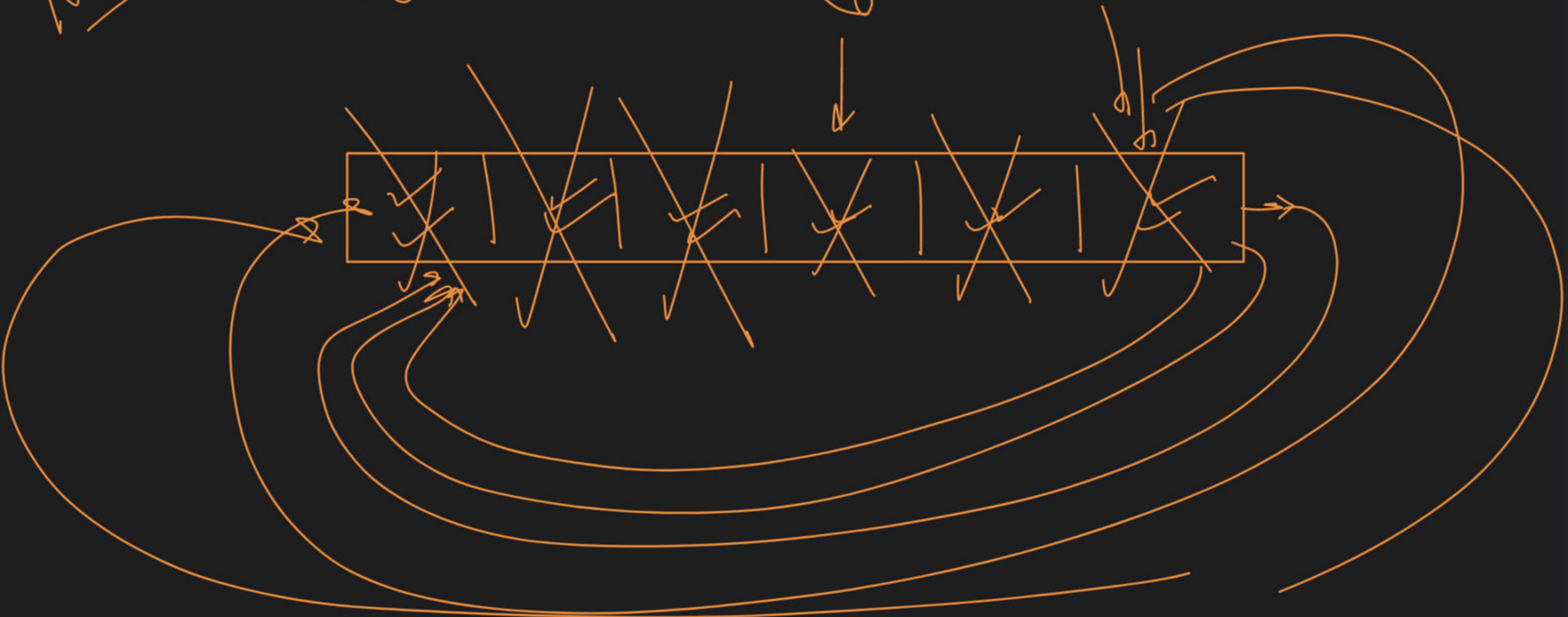
2 min

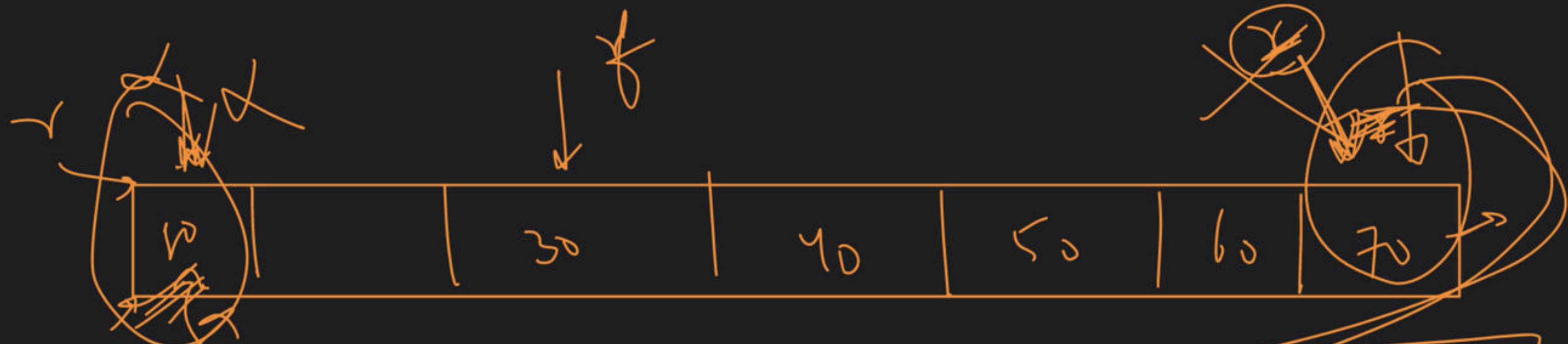
Break

→ Circular
Queue

No real
front

+ Circular
Nature





Push()

Overflow

~~$f = -1$ & $r = \text{size} - 1$~~

$r = \text{rear} = \text{front} - 1$

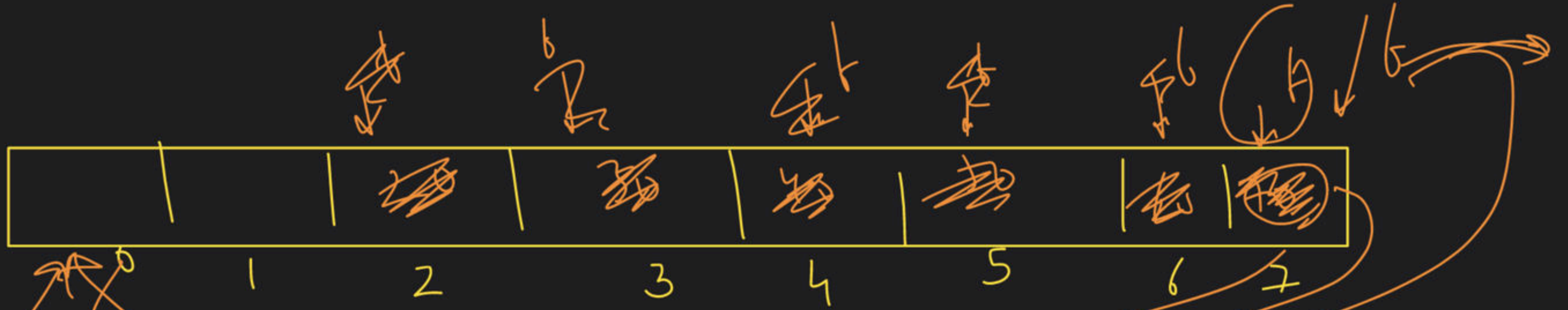
empty

$\rightarrow f++$, $r++$, $\text{arr}[r] = \text{value}$

if (circular nature $\rightarrow$ $r = \text{size} - 1$ & $f = 0$)

normal case $\rightarrow$ $\text{arr}[r] = \text{val}$

$\rightarrow r = 0$
 $\rightarrow \text{arr}[r] = \text{val}$



pop

Underflow $\rightarrow f == -1 \wedge r == -1$

single element $\rightarrow$ if (f == r) $\rightarrow f = -1, r = -1$

Circular nature $\rightarrow$ if (front == -1 || size == 1) $\rightarrow f = 0$

normal case $\rightarrow f++$

Circular doubly Ended Queue

H/W

Circular Queue

Circular
Nature

+

- push front (5)
- push back (10)
- pop front ()
- pop back ()